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Epoxide-Functionalized, Poly (Ethylenimine)- Confined Silica/Polymer Module Affording Sustainable CO₂ Capture in Rapid Thermal Swing Adsorption

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functionalization, Rapid thermal swing adsorption

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16 Creating a module that achieves sustainable CO₂ capture while being compatible with the
17 existing industry is paramount in overcoming the current CO₂-driven environmental
18 issues. This paper presents the fabrication of hollow fiber sorbents (HFSs) and their
19 respective modules to capture CO₂ by the rapid thermal swing adsorption process
20 (RTSA). Poly(amide-imide) (PAI) /microspheric SiO₂ composites were fabricated with a
21 *so-called* ‘sieve-in-a-cage’ micro-architecture which promotes CO₂ diffusion. More
22 importantly, a selected amount of 1,2-epoxybutane (EB)-functionalized
23 poly(ethylenimine) (PEI), denoted as 0.37EB-PEI, was incorporated into our PAI/SiO₂
24 composites forming PAI/SiO₂/0.37EB-PEI HFSs which enhanced the thermal stability
25 with a moderate CO₂ sorption uptake of 0.88 mmol CO₂ g⁻¹. The resulting HFSs were
26 assembled into a module using either stainless steel (SS) or poly(tetrafluoroethylene)
27 (PTFE), and their respective CO₂-capturing performances in the RTSA process were
28 compared. The two modules had a comparable breakthrough CO₂ capacity of 0.42 mmol
29 CO₂ g⁻¹ for a wet feed mixture of CO₂/He/N₂ (14 mol/ 14 mol/ 72 mol) (RH 100%) and a
30 comparable CO₂ desorption efficiency (*i.e.*, 95% desorption within 2 min.) under 100%
31 CO₂ at 120 °C due to the presence of the 0.37EB-PEI; however, different thermal
32 properties inherent to the modular materials caused the PTFE-based module to
33 outperform the SS-based counterpart in terms of cooling, enabling the execution of an
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entire RTSA cycle within 8 min. Additionally, the PAI/SiO₂/0.37EB-PEI/PTFE module maintained its breakthrough capacity of 0.42 mmol CO₂ g⁻¹ over five consecutive RTSA cycles, confirming its good long-term stability as well.

Introduction

Retrofitting an efficient CO₂-capturing process onto existing coal-fired power plants is vital to mitigate concerns about global climate change while at the same time, it would require only minor modifications of the current fossil fuel-driven industry.¹⁻³ Various processes have been developed to reduce CO₂ emissions, among which an aqueous amine-based absorption has been considered the most mature CO₂ capture technology.⁴⁻⁷ However, the high energy required for the regeneration of the aqueous amine solution with its corrosive nature necessitates the emergence of alternative CO₂ capture processes.^{1,8,9} An adsorption process based on solid amine adsorbents is an attractive alternative to the conventional absorption process because it can substantially reduce the regeneration energy due to the absence of large amount of water. Different regeneration approaches can be used including pressure swing adsorption (PSA), vacuum swing adsorption (VSA), temperature swing adsorption (TSA), or a combination of them.¹⁰ Among them, TSA seems more suitable for post-combustion CO₂ capture because both PSA and VSA have a high energy cost to compress or vacuum huge volumes of low pressure gas streams.¹¹ Besides, it can use low-grade thermal energy sources from power plants for regeneration, possibly reducing the operating cost.¹²

To further reduce the energy demand and increase the efficiency for capturing CO₂, a composite adsorbent bed-based rapid thermal swing adsorption (RTSA) process was suggested by Lively and co-workers.¹³ A full RTSA cycle is composed of four quick

1 stages; CO₂ adsorption, CO₂ desorption, gas sweeping, and system cooling with a cyclic
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3 time of less than 3 min (**Figure 1a**).¹⁴ Such rapid CO₂ adsorption/desorption kinetics can
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5 be attained by a thermal fluid passing through the bore side of the fiber sorbents with a
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7 thin wall (~300 μm). Thermal fluids have the role of a ‘shell-and-tube heat exchanger’ by
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9 carrying the heat away from the flue gas or by providing the thermal energy to the flue
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11 gas during the CO₂ adsorption or desorption stages, respectively.¹⁵ The lumen side barrier
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13 layer along the bore side is a unique characteristic of hollow fiber sorbents, which isolates
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15 the thermal fluid from the CO₂ adsorbents. Additionally, CO₂-rich flue gas fed into the
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17 shell side of the fiber sorbents can easily access the CO₂ adsorbents which are dispersed
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19 in the polymer matrix with “sieve-in-a-cage” morphology, minimizing the mass transfer
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21 resistance, and the CO₂-lean flue gas can easily be released as well (**Figure 1c**). It is
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23 noteworthy that the heat integration concept can further reduce the overall parasitic load
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25 by recovering the sensible heat of the fiber sorbents.^{16,17}

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34 Despite the many benefits stated above, engineering elements to help construct the
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36 RTSA, which is highly appealing for large-scale applications, remains somewhat
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38 unclear.¹⁸ The aforementioned elements can be summarized as follows: 1) the type of
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40 polymers, 2) the kind of amine-based CO₂ captivator confined inside the silica pores, 3)
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42 the texture of the lumen layer, and 4) the thermal properties of a module containing a
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44 bunch of the HFSs. This paper presents our efforts in improving these process elements,
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46 all of which are further specified below with their respective rationales.

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51 Initially, the RTSA process used cellulose acetate (CA) polymer/zeolite 13X as proof-
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53 of-concept HFSs for capturing CO₂.^{13,16} The CA polymer, however, was thermally
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55 unstable and mechanically weak. In addition, zeolite 13X was inappropriate as an
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‘adsorbent’ because H_2O adsorbates are preferentially adsorbed over CO_2 in a humid flue gas stream. Poly(amide-imide) (PAI) was suggested as an alternative to CA due to its enhanced thermal and mechanical stability over its CA counterpart while providing moderate hydrophilicity¹⁹ and thus served as a ‘cage’ in this study. Moreover, zeolite 13X was replaced with poly(ethylenimine) (PEI)-containing mesoporous silica particles as the CO_2 adsorbents. The sole PEI, however, was found inappropriate for long-term stability in the RTSA process due to the urea formation-induced amine deactivation at the desorption step.^{13,18,19,23} Furthermore, its high heat of adsorption causes difficulty in releasing CO_2 molecules and therefore, all the previous studies associated with the HFSs demonstrated CO_2 desorption only in an inert gas purging condition.^{19-22,24-28}

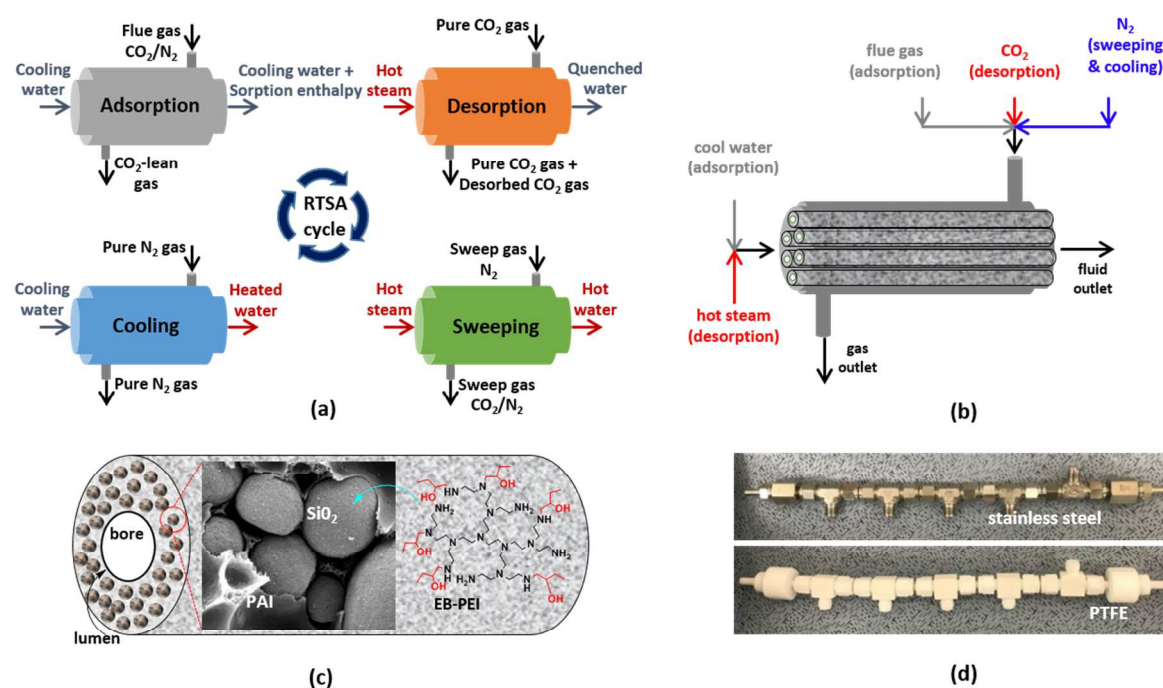


Figure 1. (a) Rapid thermal swing adsorption (RTSA) cycle. (b) Schematic illustration of RTSA process operation. (c) Hollow fiber sorbents (HFSs) comprising of silica with poly(amide-imide) (PAI), 1,2-epoxybutane-modified poly(ethylenimine) (EB-PEI), Neoprene latex lumen layer, and bore. (d) Design of a CO_2 -capturing module with the use

of two different modular materials such as stainless steel (SS) and poly(tetrafluoroethylene) (PTFE). The module also incorporates six HFSs.

Recently, Choi and co-workers functionalized PEI with 1,2-epoxybutane (EB), in which the hydroxyl groups were located near the amine functionalities.²⁹ The EB-functionalized PEI (EB-PEI) improved the long-term stability by inhibiting the formation of urea during the CO₂ desorption stage even in the pure CO₂ purging condition and promoted CO₂ sorption kinetics.³⁶ We therefore selected EB-PEI for our CO₂ adsorbents and confined them in the PAI/SiO₂ composite (**Figure 1c**). Additionally, the greatest CO₂ sorption ability of the SiO₂/EB-PEI was achieved at an optimum EB/PEI molar ratio of 10:1 (denoted as 0.37EB-PEI), which was also used in this study.²⁹ The texture of the lumen layer located between the water flow in the bore and the HFSs is another important factor to prevent an undesired mass transfer during the RTSA process.³⁰⁻³² In previous studies by Koros and co-workers, Neoprene latex (*i.e.*, poly(chloroprene)) was deposited onto the inner layer of HFSs, which formed a defect-free, impermeable lumen layer through the cross-linking of the Neoprene, and thus, it was adopted in our study to fabricate the module (**Figure 1c**).³³

In addition, it should be noted that the desorption step of the previous RTSA cyclic experiments was done with a heat tape, not by a steam, which causes difficulty in achieving fast desorption kinetics.^{20,27} We, however, performed multiple RTSA cyclic experiments consistently using a thermal fluid (H₂O) to simulate the actual operation conditions. It was also hypothesized that poly(tetrafluoroethylene) (PTFE) can outperform stainless steel (SS) as a modular material due to its lower thermal conductivity

and specific heat capacity than those of the SS. It was highly likely that compared to the SS, PTFE can not only insulate the thermal energy provided by the hot steam more efficiently during CO₂ desorption but also emit the heat accumulated in the HFSs more efficiently during the cooling process. Hence, the appropriate RTSA module was determined by comparing the CO₂ capture performances of two different modules (SS vs. PTFE) during the RTSA cycles (**Figure 1d**).

Experimental Section

Chemicals. All chemicals were used as-received, unless otherwise noted: fumed SiO₂ (KONASIL K-300; 99.8 wt%; OCI), poly(amide-imide) (PAI; Solvay Advanced Polymers), poly(vinylpyrrolidone) (PVP; M_w of 55,000; Sigma-Aldrich), poly(ethylenimine) (PEI; M_w of 1,200; Junsei), Neoprene® (solid content of 50 %; DuPont Elastomers), TSR-633 (Tiarco Chemical), N-methyl-2-pyrrolidone (NMP; extra pure, Daejung), methanol (99.8 %; Merck), n-hexane (95 %; Daejung), toluene (99.8 %; Daejung), and 1,2-epoxybutane (EB; 99 %; Sigma-Aldrich). PAI and PVP were dried under vacuum at 100 °C for 12 h to remove the moisture prior to their use.

Synthesis of SiO₂ microspheres. SiO₂ microspheres were synthesized by spray-drying of a SiO₂ slurry containing 1 wt% fumed SiO₂. In a typical synthesis, 100 g of fumed silica and 9.9 kg H₂O were mixed to form slurry, and the resultant slurry was injected into the spray dryer of the inlet and outlet were maintained at 185 and 95 °C, respectively. The resulting SiO₂ microspheres were collected using cyclone separator with a frequency of 60 Hz. These were then calcined in an air atmosphere at 800 °C for 2 h at a ramping rate of 5 °C min⁻¹ to form the three-dimensional microspheric, porous SiO₂ network.

Formation of PAI/SiO₂. PAI was hybridized with SiO₂ using a dry-jet/wet quench process to form the PAI/SiO₂ hollow fiber sorbents (HFSs) and the associated compositions and spinning conditions are shown in **Table 1**. After the spinning procedure, solvents filling the pores of the PAI/SiO₂ were first exchanged with fresh deionized water and replaced by methanol and n-hexane to minimize the collapse of the pores of the PAI/SiO₂. Then, the fibers were placed in a fume hood to air dry for 1 h and then dried for 2 h at 100 °C under a vacuum.

Table 1. Optimized spinning conditions used to fabricate the PAI/SiO₂.

parameter	condition ^a
dope composition (wt%)	61/4/5/15/15 for NMP/PVP/H ₂ O/SiO ₂ /PAI
bore composition (wt%)	88/12 for NMP/H ₂ O
extrusion flow rate (mL min ⁻¹)	5.2/1.3 for core/bore
quench temperature used during spinning (°C)	50 for H ₂ O
air gap (cm)	1

^a NMP and PVP denote N-methyl-2-pyrrolidone and poly(vinylpyrrolidone), respectively.

Formation of PAI/SiO₂/0.37EB-PEI and PAI/SiO₂/PEI. Poly(ethylenimine) (PEI) was first modified using 1,2-epoxybutane (EB) to form 0.37EB-PEI. Typically, 83.3 mmol of EB were added dropwise to the PEI (8.33 mmol) dissolved in 20 g of methanol, stirred for 12 h at 25 °C, and diluted with 124 g of methanol to form a methanol solution that consisted of 10 wt% 0.37EB-PEI. PAI/SiO₂ was then impregnated with the resulting methanol solution mixture, dried

at 25 °C for an hour, and further dried at 100 °C for 2 h. This led to the formation of the PAI/SiO₂/0.37EB-PEI. PAI/SiO₂/PEI was fabricated following the identical procedure to that for PAI/SiO₂/0.37EB-PEI except 0.37EB-PEI was replaced with PEI.

Formation of the lumen layer. The lumen layer was evolved around the bore side of the PAI/SiO₂/0.37EB-PEI and PAI/SiO₂/PEI by a post-treatment that was slightly modified based on those reported elsewhere.^{19,31,33} Neoprene[®] was used as a precursor to form the lumen layer and first mixed with TSR-633 to cross-link the Neoprene[®] at a weight ratio of 89/11. The mixed solution was fed to the bore side of six HFSs with an Isco syringe pump (Model No. of 1000D) at a flow rate of 10 mL min⁻¹, prior to which the other end of the six HFSs was immersed in n-hexane to prevent the exit of the HFSs from being clogged by the Neoprene[®]. The Neoprene[®] was swelled by a N₂ flow saturated with toluene for 3 h and then cross-linked under a vacuum at 100 °C for 2 h. The opposite side of the six HFSs was also post-treated in an identical manner to that depicted above.

Formation of the module. A lab-scale module was fabricated using the six HFSs to occupy ~45 % of the total volume provided by a cylindrical modular material, while providing the six HFSs with a CO₂-accessible longitudinal length of 27 cm. As shown in **Figure 1d**, two modular materials with an inner diameter of 0.92 cm and different thermal properties were used to fabricate the modules (*i.e.*, stainless steel (SS) vs. poly(tetrafluoroethylene) (PTFE)).

Supplementary Characterizations. N₂ isotherms of the materials were obtained at 77 K using the Micrometrics ASAP 2020 to observe their surface areas and total pore volumes via the Brunauer-Emmett-Teller (BET) theory. The ¹³C-nuclear magnetic resonance (NMR) spectra of the EB-PEI were recorded using an Agilent DD2 400 MHz NMR spectrometer operated at a ¹³C

frequency of 100.6 MHz. The element analyses (EA) of the HFSs were performed using a Flash 2000 (Thermo Scientific), whereas thermogravimetric analyses (TGA) of the HFSs were conducted with the TGA1 (Mettler Toledo). Prior to the TGA analyses, the HFS were heated at 100 °C for an hour to remove any moisture and then heated to 800 °C at a ramping rate of 10 °C min⁻¹ under an air atmosphere to determine the SiO₂ composition in the PAI/SiO₂ composites. Field emission gun-scanning electron microscopy (FEG-SEM) and energy dispersive X-ray spectroscopy (EDX) analyses of the HFSs were performed using Inspect F (FEI) at 10 kV. The CO₂ adsorption abilities of the PAI/SiO₂/0.37EB-PEI and PAI/SiO₂/PEI were evaluated with a thermogravimetric analyzer (Scinco N-1500). The material was dried at 100 °C for 3 h under a N₂ atmosphere at a flow rate of 50 mL min⁻¹ to purge the surface prior to being cooled to 35 °C. The gas was then switched to a mixed gas of CO₂/N₂ with a 15/85 molar ratio and a total flow rate of 50 mL min⁻¹, and the CO₂ capacitance of the material as a response to the CO₂ adsorption time.

CO₂ sorption measurements. The multi-cyclic CO₂ sorption abilities of the modules during the RTSA process were evaluated using a multi-component system designed and custom-built based on other previous studies (**Figure 2** and **Figure S1**).^{15,19,20,25,32} The modules were initially located inside the insulation compartment and wrapped with a heat tape (Omega Engineering) to purge the modules under a N₂ atmosphere at 120 °C for 3 h prior to the RTSA process. Throughout the RTSA recycle runs, all gases were controlled by a series of mass flow controllers and fed to the shell sides of the HFSs, whereas the total flow rate of gases was maintained at 40 mL min⁻¹ for the adsorption/desorption stages and 600 mL min⁻¹ for the sweeping/cooling stages. In addition, the thermal fluid passing through the bore sides of the HFSs was 35 °C water or 120 °C steam, each of which was supplied by Isco syringe pump (1000D) during adsorption/cooling

stages or by a steam generator (CJ Engineering) during the desorption/sweeping stages, respectively. The flow rate of the thermal fluid was 10 mL min^{-1} during adsorption/cooling stages or 5 g min^{-1} during the desorption/sweeping stages. For CO_2 adsorption, the module was fed with a wet flue gas-simulating mixture which consisted of N_2 (carrier gas), He (inert tracer), and CO_2 at a molar ratio of 72:14:14 (relative humidity of 100 %). Among the feed components, the helium gas was fed at the same partial pressure as the CO_2 gas as an inert tracer.¹⁹ For the CO_2 desorption from the HFSs, the flue gas stream was altered to a CO_2 stream. CO_2 desorption was carried out at 120°C for 2 min. The module was then subjected to a sweeping stage by changing the type of feeding gas from CO_2 to N_2 . After that, the module was cooled under a N_2 flow by introducing cooling water into the bore side. A Gaseous mixture came out from the exit of the module and was sent to mass spectrometer (Hiden HAL 201 RC) to quantify the CO_2 capturing abilities of the modules. The single-cyclic CO_2 sorption abilities of the modules during the RTSA process were also evaluated by following an identical procedure to that of the multi-cyclic CO_2 sorption experiments, unless otherwise noted in the Figure captions.

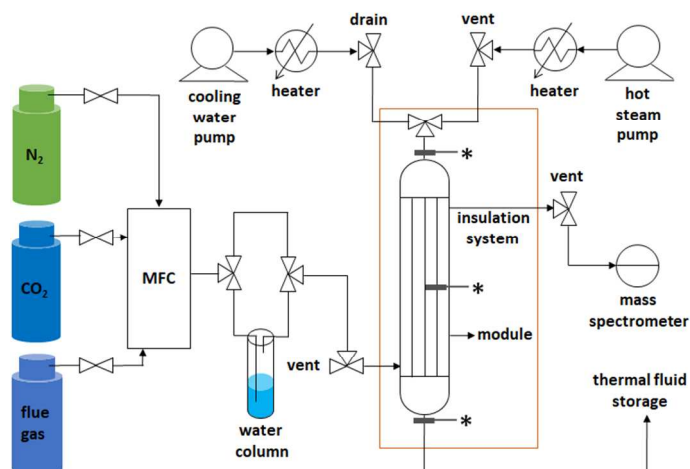


Figure 2. Multi-component system to evaluate CO₂ capturing abilities of the modules. MFC denotes mass flow controller. Asterisks (*) indicate the thermocouples to measure the temperatures for the inlet, the middle, and the outlet of the modules during the RTSA cycles.

Results and Discussion

Fabrication and characterizations of the HFSs. As specified in the experimental section, the HFSs were manufactured by four-synthetic steps: SiO₂→ PAI/SiO₂→ PAI/SiO₂/0.37EB-PEI→ lumen layer coating. The selection of our micro-spheric SiO₂ particles was based on their large porosity characterized by a total pore volume of 1.9 cm³ g⁻¹ and their excellent hydrothermal stability induced by their enhanced silica framework thickness of 10 – 15 nm.²⁹ Especially, such large pore volumes of the microspheres with a three-dimensional porous network were expected to promote CO₂ adsorption uptake as well as efficient CO₂ diffusion.

The mono-dispersed, 10 μm-sized SiO₂ sieves need to be caged with large quantity in a polymer and secured during the RTSA process to achieve a long-term stable CO₂ capturing performance. PAI was chosen as the backbone of the HFSs to support the SiO₂ particles due to its good mechanical and thermal stability induced by the hydrogen bonds between the amide and imide moieties.³⁴ Additionally, the enhanced thermal stability of the PAI/SiO₂ was validated by thermogravimetric analysis (TGA) under an air atmosphere (**Figure S2**). The high thermal resistance of the PAI showing no weight loss until 250 °C was conserved after its integration with the SiO₂ microspheres (*i.e.*, PAI/SiO₂). The TGA analysis also showed that a high quantity of SiO₂ (ca. 45 wt%) was incorporated into the PAI/SiO₂, implying that a substantial amount of 0.37EB-PEI can be accommodated. This could be potentially beneficial in capturing a large

quantity of CO₂ while attaining a significant long-term stability during the RTSA process operated at ≤ 120 °C.

We then prepared the 0.37EB-PEI to confine it into the PAI/SiO₂ for the generation of the HFS denoted as PAI/SiO₂/0.37EB-PEI via impregnation. As discussed in the introduction section, the study by Choi and co-workers demonstrated that the EB-PEI complex showed a phenomenal CO₂ sorption quantity/recyclability/kinetics, when the molar ratio of N to O included in the EB-PEI was 1:0.37.²⁹ The successful modification of the PEI by the EB was confirmed by the ¹³C-nuclear magnetic resonance (NMR) spectra and elemental analysis (EA) of the PEI and the 0.37EB-PEI (**Figure S3 and Table S1**). It is noteworthy that the functionalization of the amines with the epoxide primarily occurs on the primary amines with the least steric hindrance among all the amine functionalities, showing a substantial reduction in the primary amine contents after the epoxide functionalization (*i.e.*, 36 % $\rightarrow$ 12 %). Furthermore, the modification of the PEI by the EB via epoxidation could only increase the contents of the C, H, and O species of the resulting PAI/SiO₂/0.37EB-PEI. To validate this, PAI/SiO₂/PEI was also synthesized for comparison. As expected, lesser N contents were contained in the PAI/SiO₂/0.37EB-PEI compared to those of the PAI/SiO₂/PEI evident by the EA of both materials (*i.e.*, 2.4 vs. 3.0 mmol_N g⁻¹ in **Table 2**). A reduction in the porosities was observed for the PAI/SiO₂ after the 0.37EB-PEI confinement (*i.e.*, S_{BET} of 120 $\rightarrow$ 50 m² g⁻¹; V_{PORE} of 0.7 $\rightarrow$ 0.4 cm³ g⁻¹), which substantiated the successful infusion of the 0.37EB-PEI into the PAI/SiO₂.

Table 2. Physical and chemical Properties of the HFSs.

HFSs	S _{BET} ^a	V _{PORE} ^a	amine	N contents ^c
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	($\text{m}^2 \text{g}^{-1}$)	($\text{cm}^3 \text{g}^{-1}$)	distribution ^b (1°/2°/3° (%))	($\text{mmol}_\text{N} \text{g}^{-1}$)
PAI/SiO ₂	120	0.7	-	-
PAI/SiO ₂ /PEI	40	0.4	36/37/27	3.0
PAI/SiO ₂ /0.37EB-PEI	50	0.4	12/51/37	2.4

^a via N₂ physisorption. ^b via ¹³C-NMR analysis of PEI and 0.37EB-PEI. ^c via EA.

Field emission gun-scanning electron microscopy (FEG-SEM) analysis of the PAI/SiO₂/0.37EB-PEI was performed to demonstrate that its morphology was appropriate for efficiently capturing CO₂ during the RTSA process (**Figure 3**). The wall thickness of the HFSs was found to be ca. 300 μm , which could aid in accelerating the heat exchange between the thermal fluid and the HFSs during the RTSA process, as reported previously.¹⁸ SEM images also corroborated that a large quantity of SiO₂ microspheres (ca. 45 wt%) was evenly dispersed throughout the PAI backbone with the *so-called* ‘sieve-in-a-cage’ micro-architecture. This architecture was also reported to improve the CO₂ accessibility into the 0.37EB-PEI, while achieving rapid CO₂ sorption kinetics.^{35,36} The fast CO₂ sorption on the PAI/SiO₂/0.37EB-PEI was highly likely, as suggested by the high He permeance of ca. 43,000 GPU and the poor He/N₂ selectivity of 2.2 (**Table 3**).

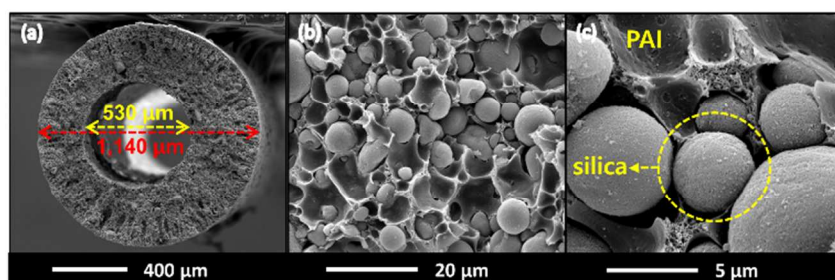


Figure 3. SEM images of the PAI/SiO₂/0.37EB-PEI: (a) cross-sectional area and (b and c) sieve-in-a-cage morphology.

Table 3. He permeances and He/N₂ selectivities for the PAI/SiO₂/0.37EB-PEI.

PAI/SiO ₂ /0.37EB-PEI	wt. % ratio of precursor solution (Neoprene [®] /TSR-633/H ₂ O)	He permeance (GPU) ^b	He/N ₂ (-)
without lumen layer	-	42,640 (±30)	2.2
with lumen layer via single cross-linking	58.9/7.1/35 ^a	32.0 (±0.2)	2.2
with lumen layer via multiple cross-linking	58.9/7.1/35 ^a	3.9 (±0.1)	2.2

^a exposed to the precursor solution twice. ^b GPU ~ X10⁻⁶ cm³ (STP) / (s·cm²·cm Hg) at 35 °C and 25 psig.

As a next step to synthesize the HFSSs that are efficient in capturing CO₂ molecules, a lumen layer was subsequently formed on the bore sides of the HFSSs to make the thermal fluid impenetrable to the PAI/SiO₂/0.37EB-PEI during the RTSA operation. As suggested by Koros and co-workers previously, we used Neoprene[®] as a precursor to form the lumen layer, and an optimized procedure was determined (**Table 3**).³³ A 10-15 μm-thick lumen layer was evolved around the bore side of the HFSSs, evident by the SEM images shown in **Figure 4a** and b. The current procedure also enabled both ends of the HFSSs to be sealed by the cross-linked Neoprene[®],

which was corroborated by the energy dispersive X-ray spectroscopy (EDX) pattern of the cross-sectional area of the HFSs based on the observation of the Cl distribution featuring the cross-linked Neoprene[®] (*i.e.*, poly(chloroprene)) (**Figure 4c**). Importantly, this procedure also led to a substantial reduction in the He permeance of the HFSs from ca. 43,000 GPU to ca. 4 GPU after the formation of the lumen layers (**Table 3**), indicating that the resulting lumen layers present on the bore sides of the HFSs were quite dense. Unfortunately, we could not completely eliminate the defects present in the lumen layers of the resulting HFSs evident by the poor He/N₂ selectivity of 2.2 even with increasing the number of the coating procedure prior to the cross-linking step. Nevertheless, the resulting lumen layer did not interfere with the multi-cyclic RTSA operation in this study.

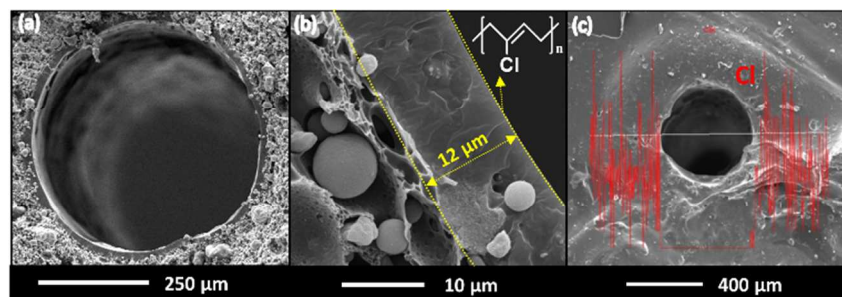


Figure 4. SEM images of the PAI/SiO₂/0.37EB-PEI with the lumen layer: (a and b) cross-sectional area and (c) EDX pattern of a cross-sectional area including the distribution of Cl species (shown as red).

Single-cyclic RTSA process: SS versus PTFE. Next, we assembled six HFSs-incorporating shell-and-tube modules to evaluate CO₂ capture performance in the RTSA process. Two different modular materials, SS and PTFE, were attempted with different thermal properties for comparison of the RTSA performance (**Table 4**). The selection of SS originated from its wide

use for the module in previous studies, whereas the selection of the PTFE was due to its smaller volumetric heat capacity and thermal conductivity than those of SS.^{19,20,25,27} It was expected that, compared to the SS, the smaller thermal conductivity of the PTFE makes it easier to insulate the thermal fluid-feeding heat energy from the outside during the CO₂ desorption stage. It was also anticipated that the module with the PTFE could emit the thermal fluid-feeding heat energy to the outside during the cooling stage more efficiently due to a potentially lower volumetric heat capacity than that of the SS. Based on the two hypotheses stated above, we expected that the PTFE module could outperform the SS counterpart during the RTSA cycle.

Table 4. Thermal properties of two modular materials.

parameter	SS	PTFE
specific heat capacity (J g ⁻¹ K ⁻¹)	0.5	1.1
volumetric heat capacity a (J cm ⁻³ K ⁻¹)	4.0	2.3
thermal conductivity (W m ⁻¹ K ⁻¹)	16.3	0.3

^a determined using specific heat capacity multiplied with density (8 g cm⁻³ for SS; 2.2 g cm⁻³ for PTFE).

We performed a single RTSA cycle following the procedure depicted in the experimental section except for some minor alterations specified in the figure captions, if any. In the case of the CO₂ adsorption stage, both the SS and PTFE modules showed comparable breakthrough CO₂ capacities with a q_b of 0.44 mmol CO₂ g⁻¹ at ca. 63 seconds and pseudo-equilibrium CO₂ uptakes with a q_{pe} of 0.88 mmol CO₂ g⁻¹ at 930 seconds (**Figure 5**). It seemed that the heat exchange between the thermal fluid (*i.e.*, cold water) and the HFSs was sufficient enough for

capturing the CO₂ molecules during the CO₂ adsorption stage regardless of the type of modular material. Both the q_b and q_{pe} , however, are smaller than those of the PAI/SiO₂/PEI(Mw; 800) fibers reported in a previous work, which had a q_b of 0.80 mmol CO₂ g⁻¹ and a q_{pe} of 1.20 mmol CO₂ g⁻¹. It is because the N contents of our fibers are smaller than those of the PAI/SiO₂/PEI(Mw; 800) (i.e., 2.40 vs. 5.90 mmol_N g⁻¹).¹⁹ We fabricated the PAI/SiO₂/PEI (800) by ourselves, and followed the identical procedure given in the reference.¹⁹ The resulting PAI/SiO₂/PEI (800) contained a N quantity of 3.4 mmol_N g⁻¹, which was smaller than that of the PAI/SiO₂/PEI (800) fabricated by Koros and co-workers (i.e., 5.9 mmol_N g⁻¹). Currently, we believe the discrepancy in the N contents inherent to the two PAI/SiO₂/PEI (800) materials could be due to the different types of SiO₂ used to synthesize them. This was conjectured based on previous studies that validated the amount of N present in CA/SiO₂/PEI materials, and the difference is mainly due to the properties of the SiO₂ such as the pore volume, BET surface area, and particle size.^{15,37} In addition, compared to our PAI/SiO₂/PEI (800), our PAI/SiO₂/PEI (1200) showed a decreased N content of 3.0 mmol_N g⁻¹. The analyses above highly suggest that the type and the appropriate selection of PEI and SiO₂ used to fabricate PAI/SiO₂/EB-PEI is important in tuning or increasing the N content for improved CO₂ adsorption. Moreover, epoxide functionalization decreased the N contents of our sorbents.³⁸

Nevertheless, our PAI/SiO₂/0.37EB-PEI modules provided better breakthrough amine efficiencies for capturing CO₂ (ca. 0.18 mmol CO₂ mmol_N⁻¹) than those of other modules reported previously (ca. 0.14 mmol CO₂ mmol_N⁻¹ for PAI/SiO₂/PEI; ca. 0.13 mmol CO₂ mmol_N⁻¹ for CA/SiO₂/PEI), suggesting that the EB functionalization enhanced the CO₂ adsorption kinetics.¹⁹ Thermogravimetric analyzer (TGA) measurements were also performed to evaluate the CO₂ uptake capacity for both PAI/SiO₂/0.37EB-PEI and PAI/SiO₂/PEI. The

PAI/SiO₂/0.37EB-PEI exhibited a lower CO₂ sorption uptake of 0.60 mmol CO₂ g⁻¹ compared to that of PAI/SiO₂/PEI (*i.e.*, 0.79 mmol CO₂ g⁻¹) due to the reduced primary and secondary amine contents (**Figure S3**). Nevertheless, the PAI/SiO₂/0.37EB-PEI could reduce the time required to uptake the CO₂ quantity at pseudo-equilibrium by one third in comparison with PAI/SiO₂/PEI. As shown in **Figure 6**, the PAI/SiO₂/0.37EB-PEI only took ca. 4,500 sec to uptake 95 % of its pseudo-equilibrium CO₂ capacitance (M_{EQ}), whereas PAI/SiO₂/PEI took ca. 12,000 sec to capture 95 % of its pseudo-equilibrium CO₂ capacitance (M_{EQ}). This result corroborated the advantage of the PAI/SiO₂/0.37EB-PEI over PAI/SiO₂/PEI for improving the amine efficiency associated with the CO₂ adsorption during the RTSA process.

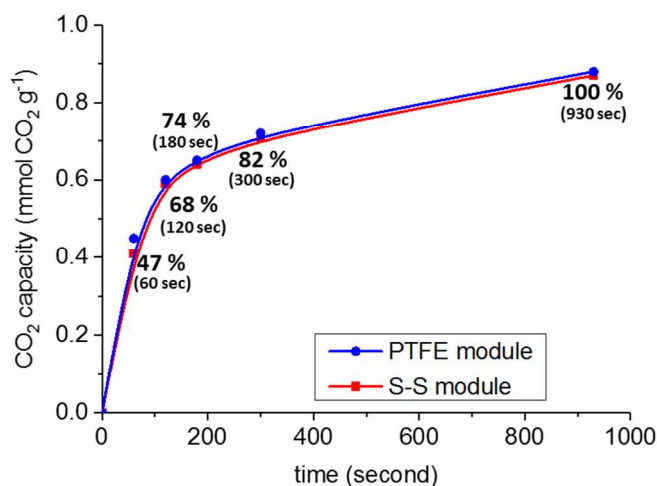


Figure 5. CO₂ adsorption capacities versus time for the modules with different modular materials such as PTFE and SS. CO₂ adsorption conditions: humid flue gas of N₂/He/CO₂ of 72:14:14 (molar ratio) with 40 mL min⁻¹; thermal fluid of water at 35 °C with 10 mL min⁻¹; 1 atm.

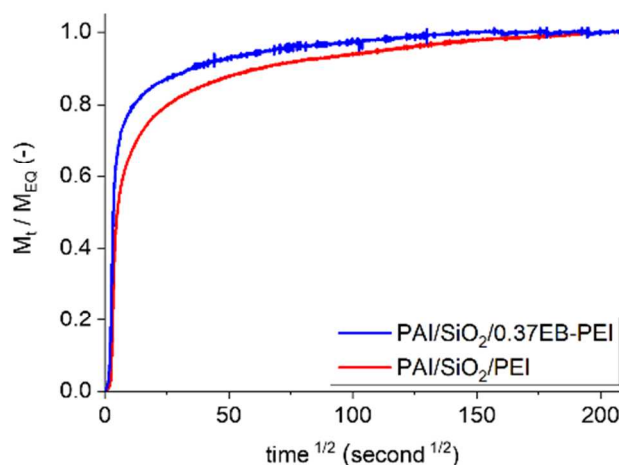


Figure 6. CO₂ adsorption abilities (M_t/M_{EQ} (-) versus square root of time) for PAI/SiO₂/0.37EB-PEI and PAI/SiO₂/PEI. M_t and M_{EQ} stand for CO₂ capacitance at the specific time (t) and pseudo-equilibrium CO₂ capacitance, respectively. CO₂ adsorption conditions: mixed gas of CO₂/N₂ of 15:85 (molar ratio) with 50 mL min⁻¹; 35 °C; 1 atm.

Although the replacement of the PEI with 0.37EB-PEI could circumvent the required use of pure N₂ purging gas for desorbing the CO₂ species, pure N₂ was used to detach the CO₂ for the single-cyclic RTSA process. This was due to the necessity of characterizing the CO₂ desorption rates for the two different modules to demonstrate the potential advantage of the PTFE over the SS as a modular material. At the initial CO₂ desorption stage (*i.e.*, ≤ 40 second), the CO₂ desorption rate was indeed faster in the PTFE-based module than in the SS counterpart shown in **Figure 7**. This could partially support our claim that the use of the PTFE modular material could help decrease the loss of heat energy conveyed by the hot steam more proficiently than by that of the SS counterpart. However, the CO₂ desorption rates for the PTFE-based module continuously declined after 40 seconds and showed an identical amount of CO₂ desorbed from the HFSs to that obtained in the SS-based module after 80 seconds. However, 95 % of the captured CO₂

species was successfully desorbed from the HFSs at ca. 120 seconds. This result suggests that both modules could provide a benefit to shorten the time required for completing the RTSA cycle.

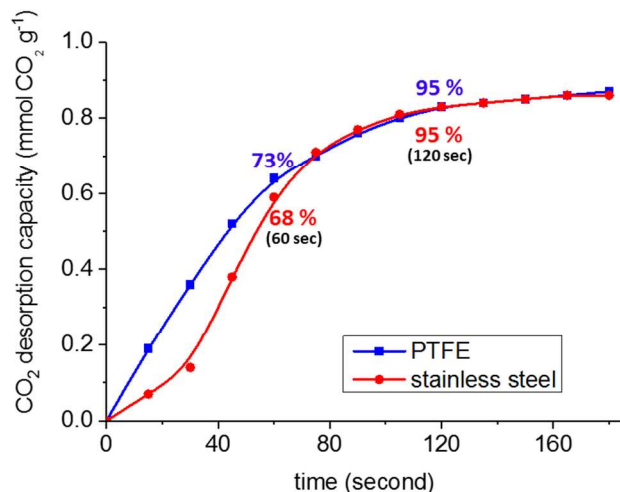


Figure 7. CO₂ desorption capacities versus time for the modules with different modular materials such as PTFE and SS. CO₂ desorption conditions: N₂ with 200 mL min⁻¹; thermal fluid of hot steam at 120 °C with 5 g min⁻¹; 1 atm.

After the sweeping stages with no discernable character observed between the SS and PTFE modules, their cooling stages were initiated by altering the type of thermal fluid from 120 °C steam to 35 °C water (See the change in the inlet temperature in **Figure 8**). The cooling performance of each module was evaluated by acquiring the temperature profiles at the middle and the outlet of the modules. This was because these profiles were helpful in verifying our hypothesis that the ‘greater ability of the PTFE to emit the heat toward the outside is attributed to its lower volumetric heat capacity than that of the SS during the cooling stage.’ Prior to changing

the type of thermal fluid for cooling, the temperature of the PTFE-based module was observed to be slightly greater than that of the SS counterpart (See the two dotted lines in **Figure 8**). This was anticipated due to the lower thermal conductivity of the PTFE than that of the SS, leading to accumulation of a greater quantity of heat in the PTFE-based module than in the SS-based counterpart during the sweeping stage. Interestingly, the SS-based module retained its bulk temperature at 50-55 °C up to 120 seconds and barely reached 35 °C after 1080 seconds. This was in contrast to the PTFE-based module exhibiting an abrupt reduction in the initial bulk temperature (70 °C) to 35 °C only within 120 seconds. This was co-current to the temperature profiles for both modules at the outlet: The outlet temperature of the PTFE-based module decreased to 35 °C within ca. 120 sec, whereas the SS-based module had an outlet temperature > 50 °C even after ca. 120 sec (see the two solid lines in **Figure 8**). These results served as evidence to support our claim and justify the use of the PTFE-based module to shorten the total processing time needed to terminate the single RTSA cycle.

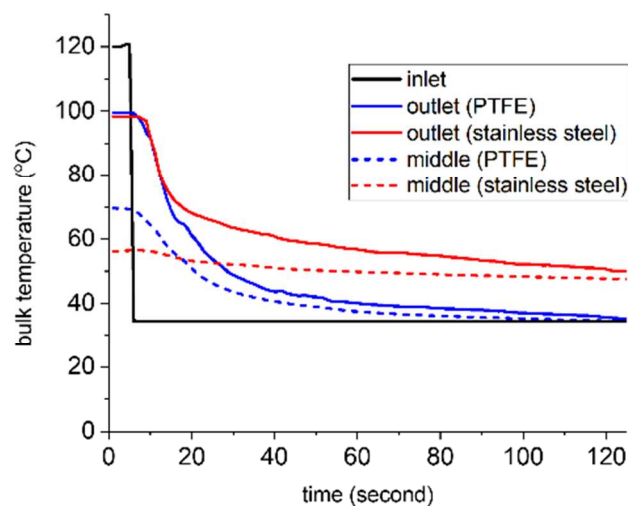


Figure 8. Bulk temperature profiles recorded at various locations of the modules during the cooling stage: the inlet (solid black), the middle (dotted blue for PTFE; dotted red for SS), and

the outlet (solid blue for PTFE; solid red for SS). Cooling conditions: N₂ with 200 mL min⁻¹; thermal fluid of water at 35 °C with 10 mL min⁻¹; 1 atm.

Multi-cyclic RTSA process: longevity and CO₂ concentration of the product stream. The long-term stability of the PTFE-based module (*i.e.*, PAI/SiO₂/0.37EB-PEI/PTFE) was evaluated by performing multi-cyclic RTSA processes (**Figure 9**). Each 2 min stage was designed based on the consideration of the times to achieve good breakthrough CO₂ capacity (q_b) for adsorption, 95 % detachment of the adsorbed CO₂ species for desorption, ≤ 1 % of residual CO₂ species for sweeping, and ≤ 35 °C as the bulk modular temperature for cooling. PAI/SiO₂/PEI/PTFE was also fabricated and its long-term stability was evaluated for comparison. The PAI/SiO₂/0.37EB-PEI/PTFE exhibited excellent longevity regarding adsorption/desorption of CO₂ molecules, as is evident from the retention of the q_b as 0.42 mmol CO₂ g⁻¹ over the five cycles. In contrast, the PAI/SiO₂/PEI/PTFE had a greater q_b (0.68 mmol CO₂ g⁻¹) than that of the PAI/SiO₂/0.37EB-PEI/PTFE in the 1st cycle due to the larger amounts of amines loaded per unit mass of the HFSs; however, its q_b continually decreased and became smaller than that of the PAI/SiO₂/0.37EB-PEI/PTFE at the 5th cycle most likely due to the urea formation-induced amine deactivation during the CO₂ desorption steps.²⁹ **Table S2** shows a comparison of our sorption capacity with those of previous works.

The CO₂ concentration in the product stream is also a critical parameter for the performance evaluation of the RTSA process. All the previous works associated with fiber sorbents have used N₂ gas for the desorption process to avoid the deactivation of the amine groups induced by the occurrence of urea formation when the amines are exposed to CO₂ at high temperature.²⁰ Fan *et al.* confirmed that an approximately 50% purity of CO₂ gas was recovered by using N₂ gas in a continuous stirred tank desorption mode.²⁷ Rezaei *et al.* reported that the CO₂ gas can be

recovered at a purity of greater than 90% by using N₂ gas in a plug flow mode based on their modeling study.¹⁴ Because our PAI/SiO₂/0.37EB-PEI hollow fiber sorbents enabled desorption in a pure CO₂ desorption condition, our product stream is expected to generate a high CO₂ purity greater than 90% by using the CO₂ gas in a plug flow mode. Of note, the CO₂ adsorption stage was conducted in the presence of H₂O, which requires to dehumidify wet, concentrated CO₂ species desorbed from the surface of HFSs.^{20,39}

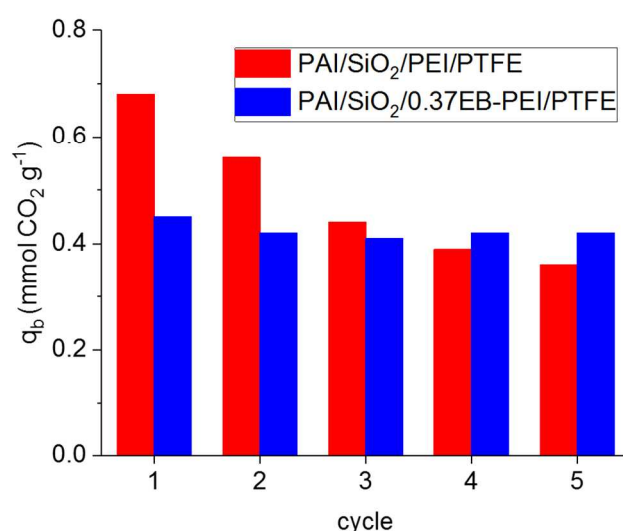


Figure 9. Breakthrough CO₂ capacity (q_b) for PAI/SiO₂/PEI/PTFE and PAI/SiO₂/0.37EB-PEI/PTFE obtained during the multi-cyclic RTSA process.

In summary, we built modules that capture CO₂ by rapid thermal swing adsorption (RTSA) operated under conditions simulating the real process. Several engineering elements that can affect CO₂-capturing performance of the modules were carefully considered and chosen prior to the construction of the modules. Poly(amide-imide) (PAI) was selected as a cage to rigidly support mono-dispersed, micro-spheric SiO₂ sieves and the PAI/SiO₂ led to facilitated permeation of a flue gas during the CO₂ adsorption/desorption stages. Epoxide-functionalized

poly(ethylenimine) (0.37EB-PEI) also served as an amine supplier to reversibly and efficiently bind with CO₂ through the confinement of the 0.37EB-PEI into the pores of the SiO₂. The resulting hollow fiber sorbents (HFSs: PAI/SiO₂/0.37EB-PEI) exhibited excellent qualities as a CO₂-capturing medium such as an increased thermochemical stability, a moderate CO₂ adsorption quantity, and fast kinetics/recyclability for multiple CO₂ sorptions. The aforementioned advantages were substantiated to be more pronounced, when poly(tetrafluoroethylene) (PTFE) was used as a modular material to assemble the HFSs. Moreover, the use of PTFE also provided the additional benefit of attenuating the time required to complete a single RTSA cycle within 8 min due to the thermal properties of the PTFE.

ASSOCIATED CONTENT

Supporting Information. The Supporting Information (multi-component system, TGA profiles of PAI and PAI/SiO₂, ¹³C NMR spectra and elemental analysis for PEI and 0.37EB-PEI, and q_b from previous researches) is available free of charge on the ACS Publications website at DOI: 10.1021/acssuschemeng.

an equipment custom-built for RTSA process, TGA analyses of PAI and PAI/SiO₂, ¹³C-NMR analyses of PEI and 0.37EB-PEI, EA analyses of PEI and 0.37EB-PEI, comparison of q_b in the literature reports.

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Author Contributions

⊥ S. Park and J. Kim are contributed equally to this work.

Notes

The authors declare no competing financial interest.

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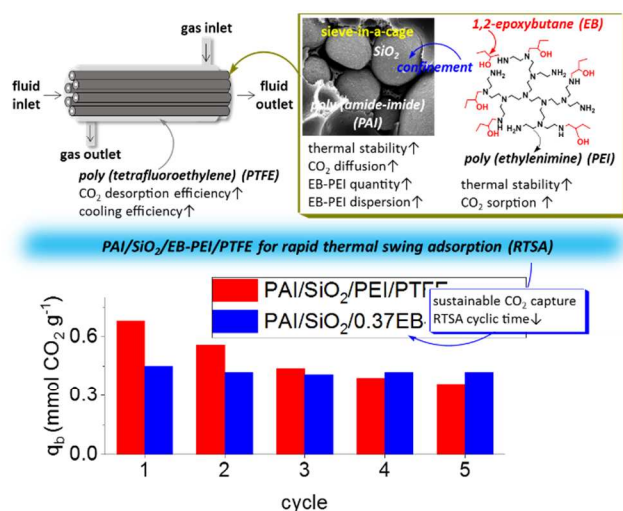
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TOC/Abstract Graphic



SYNOPSIS

Poly(amide-imide)/SiO₂ composites in a poly(tetrafluoroethylene) module confined 1,2-epoxybutane-functionalized poly(ethylenimine) and showed moderate, longevous CO₂ capture via rapid thermal swing adsorption.